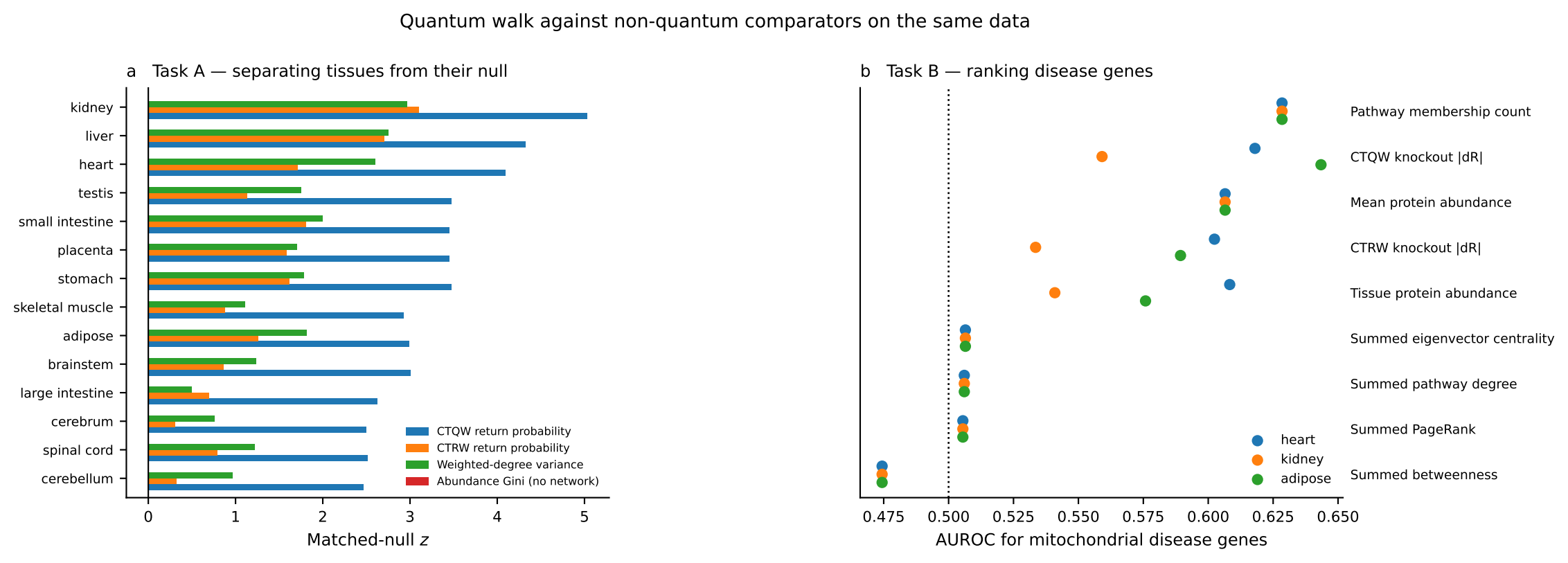
Quantum walk against non-quantum comparators on the same data
a Task A — separating tissues from their null
kidney
liver
heart
testis
small intestine
placenta
stomach
skeletal muscle
adipose
brainstem
large intestine
cerebrum
spinal cord
cerebellum
0
1
2
3
4
5
Matched-null z
CTQW return probability
CTRW return probability
Weighted-degree variance
Abundance Gini (no network)
b Task B — ranking disease genes
Pathway membership count
CTQW knockout |dR|
Mean protein abundance
CTRW knockout |dR|
Tissue protein abundance
Summed eigenvector centrality
Summed pathway degree
Summed PageRank
Summed betweenness
heart
kidney
adipose
0.475
0.500
0.525
0.550
0.575
0.600
0.625
0.650
AUROC for mitochondrial disease genes